negotiation and risk management; Requirements analysis and specification - modeling techniques: inception vs. specification, techniques for writing high-quality requirements, documentation standards, UML notations, external qualities management, contract specification; Requirements verification, and validation: detection of conflicts and inconsistencies, completeness, techniques for inspection, feature interaction analysis and resolution; Requirements management: traceability, priorities, changes, baselines, tool support; Examples of requirements for various types of systems: embedded systems, consumer systems, web-based systems, business systems; requirements engineering in RUP, requirements engineering in agile methods.

**Recommended Texts:**

1. R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. McGraw-Hill, 2019
2. I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2015

**CSE 4404** **Algorithms Lab** **Credit: 1.00**

Sessional works based on CSE 4403.

**CSE 4410** **Database Management Systems II Lab** **Credit: 1.50**

Sessional works based on CSE 4409.

**CSE 4412** **Data Communication and Networking Lab** **Credit: 1.00**

Sessional works based on CSE 4411.

**SWE 4402** **Software Requirement and Specifications Lab** **Credit: 1.00**

Sessional works based on SWE 4401.

**SWE 4404** **Software Project Lab II** **Credit: 1.50**

Each student will be assigned a single project. It will test the ability of the students to handle large projects. Students will focus on developing web based, networked, and mobile applications. The students have to follow formal methods of system analysis and software development processes. They must familiarize themselves with standard version control and software development environments such as Github, IntelliJ etc.



# Fifth Semester

**Math 4543** **Numerical Methods** **Credit: 3.00**

Solution of algebraic and Transcendental equation: Iterative method, Gauss elimination method, Gauss-Seidel method and their applications in Engineering fields.

Interpolation/Extrapolation: Interpolation with one and two independent variables. Formation of different difference table. Newton's forward and backward difference, Lagrange's interpolation, Neville-Aitken's interpolation, Successive iteration.

Numerical Integration: Trapezoidal rule, Gauss's Quadratic formula, Multiple integration, Romberg's method, Truncation and error estimation. Numerical solution of differential equations ,Numerical solution of partial differential equations, curve fitting, Methods of least square, Estimation of linear and nonlinear parameters, formulation, different engineering experimental results.

**Recommended Texts:**

1. J. D. Faires and R. L. Burden, *Numerical Methods*, 4th ed. Cengage Learning, 2012
2. M. A. Celia and W. G. Gray, *Numerical Methods for Differential Equations: Fundamental Concepts for Scientific and Engineering Applications*, 1st ed. Pearson College Div, 1991
3. L. W. Johnson and R. D. Riess, *Numerical Analysis*, 2nd ed. Addison-Wesley Publishing Company, 1982

**CSE 4501** **Operating Systems** **Credit: 3.00**

Types of operating systems: single user, real-time, batch, multiple access. Principles of operating systems; design objectives; sequential processes; concurrent processes, concurrency, functional mutual exclusion, processor co-operation and deadlocks, processor management. Control and scheduling of large information processing systems.

Resource allocation, dispatching, processor access methods, job control languages. Memory management, memory addressing, paging and store multiplexing. Multiprocessing and time sharing, batch processing. Scheduling algorithms, file systems, protection and security; design and implementation methodology, performance evaluations and case studies.

**Recommended Texts:**

1. A. Silberschatz et al., *Operating System Concepts*, 10th ed. Wiley, 2018
2. A. S. Tanenbaum, *Modern Operating Systems*, 5th ed. Pearson Education, Inc., 2022